

Since $KE_A = KE_B$, then $GPE_A = GPE_B$ and $TE_A = TE_B$. Hence B is false.

- 14** Fig. 14.1 shows the variation of velocity v with displacement x of a simple harmonic oscillator. The state of motion of point P is shown in Fig. 14.1.

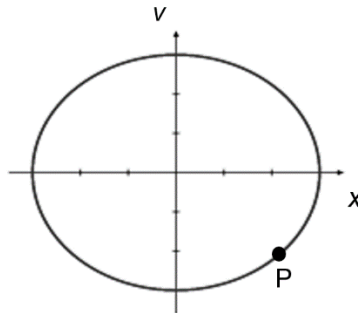


Fig. 14.1

Fig. 14.2 shows the variation with time t of the acceleration a of the oscillator.

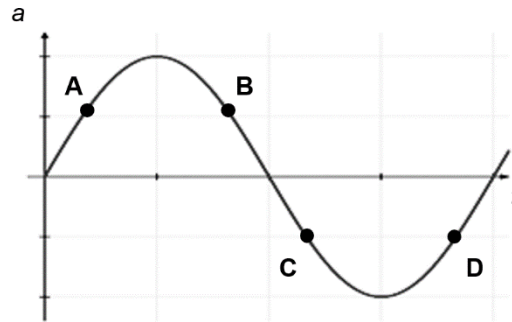


Fig. 14.2

Which of the points on Fig. 14.2 correspond to the state of motion represented by point P?

Ans: D

At point P, the displacement is positive and velocity is negative.

From the definition of SHM, acceleration and displacement are always oppositely directed. Thus, a should be negative. $\rightarrow$ A and B not possible.

For v , sketch $x-t$ graph and determine whether point C or D has negative velocity (i.e. where gradient of $x-t$ graph is negative). (Using $a = -\omega^2 x$, since $a-t$ graph is a positive sine graph, $x-t$ graph is a negative sine graph.)